

RFID Attendance Logger

Power, Capacity and Cost

Current draw for each operating mode and event, when the MCU and the reader are awake, the reasoning behind each clock frequency, the expected number of card reads and the battery life on a 1000 mAh, 3.7 V Li-ion cell, how the STM32L432KB's 128 kB of flash is shared between the program and the attendance log, and what a finished unit costs to make at 1 to 10 000 units, in US dollars and Sri Lankan rupees.

Hardware	Battery	Date	Status
STM32L432KB (128 kB) + ST25R3916	1000 mAh, 3.7 V nominal (3.7 Wh)	08 October 2026	Currents measured 8 October 2026

BATTERY LIFE

18 months

4 lectures/day, 100 students, incl. self-discharge (13 pessimistic)

IDLE SCANNING

194 μA

Reader wake-up mode is 95 % of it

SWITCHED OFF

1.8 μA

Standby, measured; 57 years on the device alone

PER CARD TAP

2.5 μAh

8.9 mC; 6,858 records fit in flash

1 Summary

The logger spends nearly all of its life in one of two low-power states. Switched off, it sits in Standby and draws **1.8 μA** (measured). Switched on and waiting for cards, the MCU sleeps in Stop 2 and wakes only for a 10 ms heartbeat flash every 5 s, while the ST25R3916 checks the antenna itself every 100 ms. That idle state draws about **194 μA** , and 95 % of it is the reader's wake-up mode. The large currents (203 mA for the RF field, 56.5 mA for the motor) flow only for milliseconds per tap, so a whole 100-student lecture costs about **0.34 mAh** (1.26 mWh).

At four lectures a day, the device uses about **31 mAh a month**. With the cell's self-discharge (about 20 mAh a month) that gives roughly **18 months per charge**, or 29 months from the device's draw alone.

The program and the log share the STM32L432KB's 128 kB flash: 72 kB for the program (60.6 kB at -Og) and 56 kB for the log, **6,858 records** or about 64 lectures of 100 students. The log fills long before the battery empties (in about 16 teaching days at four lectures a day), so the PC sync, which clears the log, sets the routine, not the charger.

A finished unit costs about **US\$25 (LKR 8,382)** to make at 1 000 units and US\$18 (LKR 6,042) at 10 000, landed in Sri Lanka and tested (section 11).

Confidence. Every current in this report except the 24/80 MHz run currents, the ADC and the flash was measured on the board (8 October 2026) with a multimeter in series with the cell, using a power-test build that holds each load steady (section 3). Event charges are those currents multiplied by the firmware's timings, which come from its constants, not from a scope. The reader's wake-up mode average is the least certain figure: the meter read 117-204 μA while it averaged the reader's 100 ms measurements, so the pessimistic column uses 204 μA and 4 false wake-ups a minute. The RF field current is for the untuned antenna.

2 Clock frequency selection

Each clock was picked for the job it does. The scanning clock matters most, because it runs on every wake-up.

Clock domain	Selection	Setting	Why
Scanning (SYSCLK/HCLK)	MSI range 6, 4 MHz	Range 1, 0 wait states	Reset clock and the Stop 2 wake-up clock (STOPWUCK = MSI): no PLL to relock on each wake-up. The work is time-bound (5 ms field guard, 106 kbit/s frames, ADC sampling), so a faster clock only raises the current for the same window.
USB session	MSI range 9, 24 MHz	Range 1, 1 wait state	USB FS needs HCLK above 14.2 MHz (RM0394). 24 MHz is the first MSI step with margin for rendering CSV sectors on demand. Powered from VBUS, so it costs the battery nothing.
USB kernel clock	HSI48 trimmed by CRS	Locks to the host's 1 kHz SOF	Meets the USB $\pm 0.25\%$ tolerance with no crystal and no PLL. Stopped outside USB sessions.
RTC and Stop 2 timer	LSE 32.768 kHz	RTC; LPTIM1 at LSE / 8 (16-bit)	Runs through Stop 2 for well under 1 μA . LPTIM1 counts 4096 Hz, so its 16-bit count wraps every 16 s and one Stop 2 sleep can last up to 15 s (the loop asks for at most 5 s); resolution 244 μs . A wake from the wrap alone goes straight back to Stop 2.
Reader SPI	4 MHz / 8 = 500 kHz	Mode 1, ST25R3916 max 10 MHz	Well inside the chip's limit. Register traffic during a read adds an estimated 2-3 ms of field-on time; a /2 prescaler (2 MHz) would cut it to under 1 ms.
ADC	HCLK / 4 = 1 MHz	640.5-cycle sample	0.64 ms per channel gives the 1.7 M Ω divider time to settle; the ADC is powered only for each sample.
Reader crystal	27.12 MHz	2 $\times$ 13.56 MHz carrier	Fixed by ISO 14443 ($f_c = 13.56\text{ MHz}$). Runs only in Ready mode or for each wake-up measurement.

Why 4 MHz and not faster

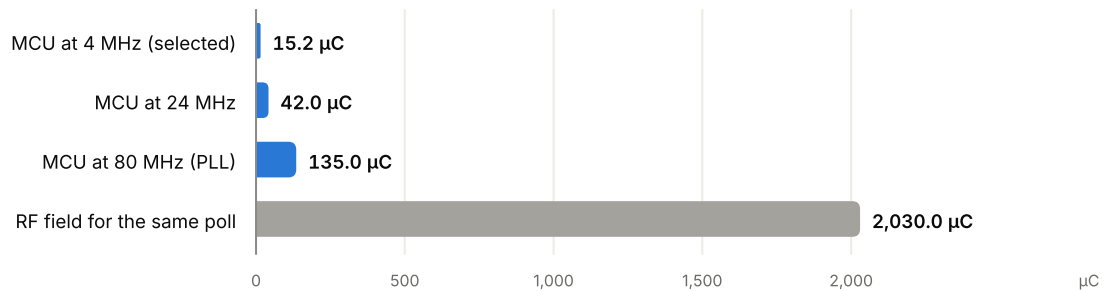
The MCU's awake time is set by the radio and the protocol rather than by computation: a read waits out a 5 ms field guard, exchanges ISO 14443 frames at 106 kbit/s, and busy-waits on the reader's IRQ line. For a fixed

window t , the charge is $Q = I_{run}(f) \times t$, so the lowest clock that meets the deadlines wins.

MCU charge during one 15 ms read window (4 MHz measured, 24/80 MHz datasheet):

4 MHz	=	1.01 mA × 15 ms	=	15.2 μC
24 MHz	=	2.8 mA × 15 ms	=	42.0 μC
80 MHz	=	9.0 mA × 15 ms	=	135.0 μC + PLL lock per wake-up
RF field	=	203 mA × 10 ms	=	2030 μC

Charge for one card read: MCU clock options against the RF field



At 4 MHz the MCU is 0.7 % of a read's charge. At 80 MHz it would be about 7 %, for no gain in speed.

For purely computational work (one pass of the main loop is roughly 2000 cycles), current per MHz is close to constant, so finishing faster and sleeping longer gains little: 0.5 ms at 1.0 mA (0.51 μC) at 4 MHz against 25 μs at 9 mA (0.23 μC) at 80 MHz. That small gain is lost to the PLL's start-up time and to the 4 flash wait states 80 MHz needs.

Why not slower

- **SPI with the field on.** The SPI clock is $\text{SYSCLK} / 8$. At 2 MHz the reader's register traffic would take twice as long while the 203 mA field is on, costing more than the MCU saves.
- **ADC sampling.** The ADC clock is $\text{HCLK} / 4$. A slower clock stretches the 640.5-cycle sample and keeps the ADC powered longer.
- **Low-power run (at most 2 MHz) would barely help.** The MCU is awake about 0.02 % of the time in idle, so its run current adds only ~0.2 μA to the 194 μA idle figure.

Voltage scaling

The firmware keeps Range 1 at all clocks. Range 2 would cut run current by about 10 %, but that saves well under 0.1 μA in idle, while USB needs Range 1 anyway. Staying in Range 1 avoids switching the regulator on every USB attach. Stop 2 and Standby use the low-power regulator either way.

3 Measurements and assumptions

Battery-side currents, measured on 8 October 2026 with a multimeter in series with the cell. The power-test build (make power-test, Bsp/Src/bsp_power_test.c) runs no application: each press of the button holds the next load steady, so the meter reads it instead of a millisecond pulse. The 3.3 V rail comes from a linear LDO, so the battery supplies the same current the MCU and LEDs draw; the reader's VDD/VDD_TX and the motor sit straight on BAT+.

Measured **MEASURED 8 OCTOBER 2026**

Mode	State held	Current	Note
1	MCU Stop 2 + RTC + LPTIM1, reader power-down	3.3 μ A	Includes the divider, LDO and charger leakage
2	As 1, reader in wake-up mode (100 ms)	188 μ A	Meter read 117-204 μ A as it averaged the 100 ms measurements; 188 averaged
3	As 1, reader Ready (oscillator on, field off)	5.66 mA	Reader between polls and during patterns
4	As 1, RF field on	203 mA	Antenna not yet tuned
5	MCU Run, MSI 4 MHz, reader power-down	1.01 mA	DS11451 gives ~0.5 mA; peripheral clocks are on
6	MCU Sleep, MSI 4 MHz, SysTick every 1 ms	0.72 mA	The loop's mode in USB sessions
7	Green LED on (470 Ω from 3.3 V)	2.77 mA	
8	Red LED on	3.0 mA	
9	Vibration motor on (coin ERM on BAT+)	56.5 mA	
10	Standby (switched off)	1.8 μ A	4.8 μ A before the reader's SPI pins were pulled down in Standby

Changes made after the first measurements

Change	Before	After
Debug clocks off in Stop 2 / Standby (DBGMCU_CR cleared at boot)	0.3 mA switched off	4.8 μ A
Reader SPI, LED and motor pins pulled down in Standby	4.8 μ A switched off	1.8 μ A
Heartbeat flash	30 ms every 4 s	10 ms every 5 s
Longest idle sleep (APP_SLEEP_MAX_MS); LPTIM1 / 8	1 s (1.35 wakes/s)	5 s (0.4 wakes/s)
Battery sample placed on the heartbeat's wake	own wake every 10 s	none extra
Empty polls before re-arming (APP_NFC_REMOVE_MISSES)	3	1
Accepted-card buzz (APP_FB_ACCEPT_VIB_MS)	90 ms	60 ms

Still estimated

Quantity	Typical	Pessimistic	Source / basis
STM32L432 Run, 24 MHz / 80 MHz	2.8 / 9.0 mA	-	DS11451 typ., Range 1 (not used while scanning)
ADC on, extra over Run	0.2 mA	-	DS11451
Flash program / erase	7.0 mA	-	DS11451; 22 ms per page erase
Button pull-ups (R8 10k + internal)	0.41 mA	-	Only while pressed
Field time per read / per empty poll	10 / 6 ms	14 / 8 ms	5 ms guard + 1 ms tick, then REQA..SELECT at 106 kbit/s
Reader wake-up mode average	185 µA	201 µA	Measured range: 117 - 204 µA on the meter
False wake-ups while scanning	1 / min	4 / min	Not measured yet; read dbg_nfc_false_wakes
Cell self-discharge	2 %/month	3 %/month	Typical Li-ion at room temperature
Usable capacity of 1000 mAh	900 mAh	850 mAh	Cut-off at 3.3 V, ageing margin

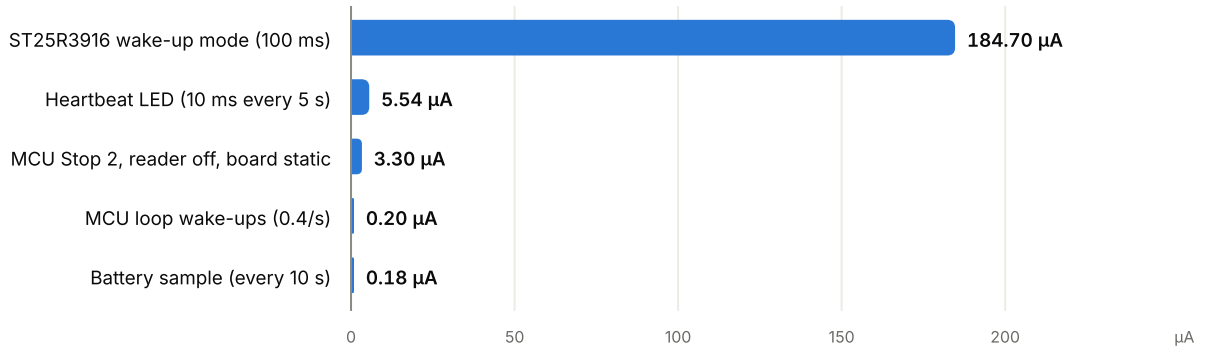
Measured on the board MEASURED 07 OCTOBER 2026

Quantity	Value	Note
Flash size (0x1FFF75E0)	128 kB	Fitted part is the STM32L432KB
Cell voltage (dbg_battery_mv)	4,152 mV	Three samples 10 s apart: 4152, 4149, 4154 mV
State of charge (dbg_battery_percent)	96 %	From the firmware's Li-ion voltage curve
ADC counts at PA7	1,876	$PA7 = 3308 \text{ mV} \times 1876 / 4095 = 1515 \text{ mV}$
VDDA from VREFINT	3,308 mV	$VREFINT_CAL \ 1652 / 1497 \text{ counts} \times 3000 \text{ mV}$
Divider check	4,153 mV	$1515 \text{ mV} \times (4.7 \text{ M} + 2.7 \text{ M}) / 2.7 \text{ M}$

4 Operating modes

Mode	Typical	Pessimistic	What is running
Off (Standby)	1.8 µA	1.8 µA	MCU Standby + RTC, reader power-down, divider (measured)
Idle, scanning	194 µA	210 µA	Stop 2, reader wake-up mode, heartbeat
Polling mode (wake-up off)	18.2 mA	-	Reader Ready + field every 100 ms; for comparison
Card read (RF field on)	203 mA	-	For ~10 ms per read; antenna untuned (measured)
Vibration motor on	56.5 mA	-	60 - 250 ms per pattern (measured)
USB session	4 - 6 mA	-	From VBUS, not the battery (plus charge current)

Idle scanning current, 194 µA in total



The reader's own antenna measurements are 95 % of idle current. The MCU, including its wake-ups, is under 3 %.

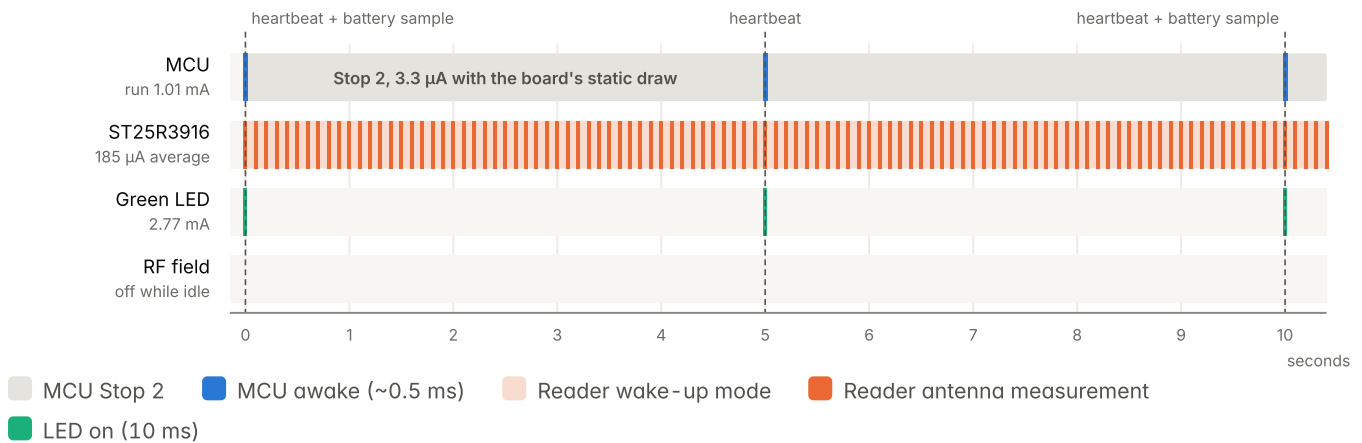
Wake-up mode	= mode 2 - mode 1 = 188 - 3.3 µA	= 184.7 µA
Heartbeat	= 2.77 mA × 10 ms / 5000 ms	= 5.5 µA
Wake-ups	= 0.4 /s × 0.5 ms × 1.01 mA	= 0.20 µA
Sampling	= 1.5 ms × (1.01 + 0.2) mA / 10 s	= 0.18 µA
Off	= measured, mode 10	= 1.80 µA
Polling mode	= 5.66 mA Ready + 203 mA × 6 ms / 100 ms + MCU	= 18.2 mA

The reader's wake-up mode is still far cheaper than polling: switching it off would take idle current from 194 µA to about 18 mA, and battery life left switched on from 193 days to 2.1 days. At 100 ms it costs about 18 µC per measurement, far more than the ~1 µC first assumed: each measurement starts the 27.12 MHz oscillator and drives the untuned antenna. A longer period (section 10) or a tuned antenna is the largest remaining saving.

5 When the MCU and the reader are awake

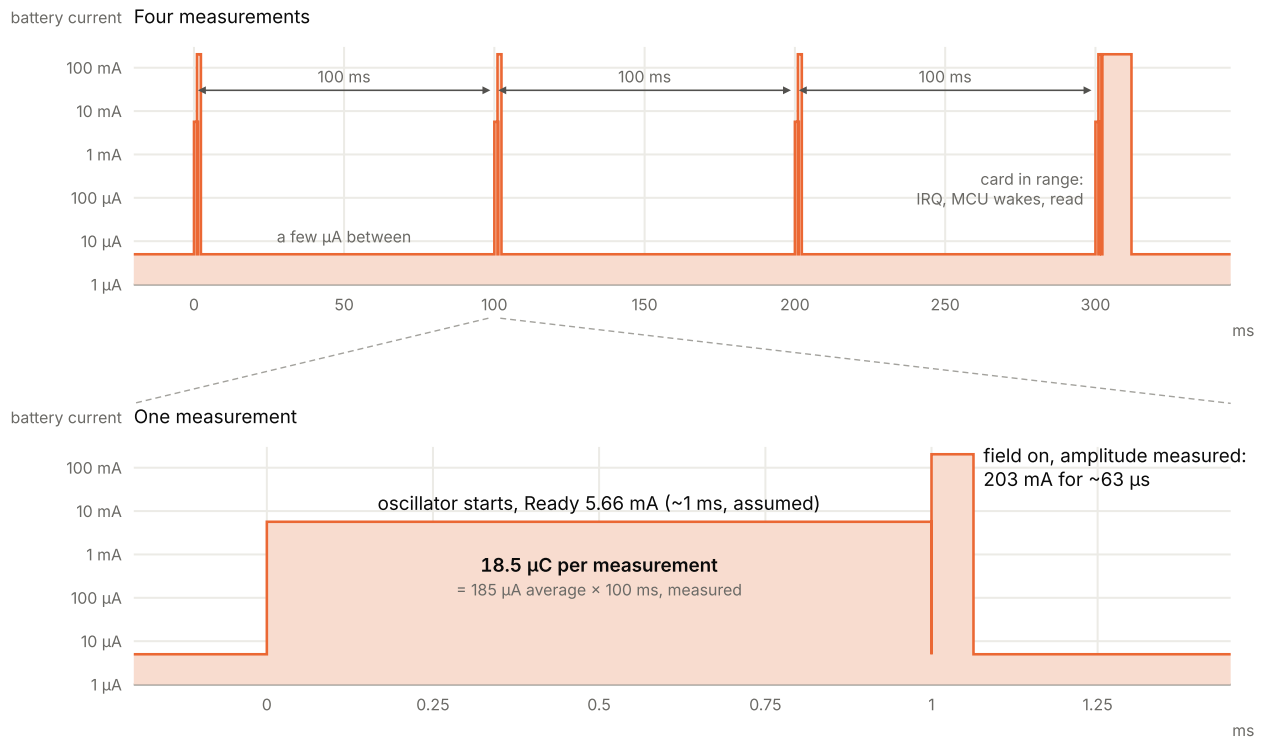
The two chips sleep independently. The MCU sleeps in Stop 2 and wakes on its own timer (LPTIM1) or on an interrupt: the button, VBUS, or the reader's IRQ line. The ST25R3916 runs its own wake-up timer and wakes the MCU only when the antenna changes. Timings are the firmware's; the currents beside each lane are the measured ones.

Idle scanning: ten seconds



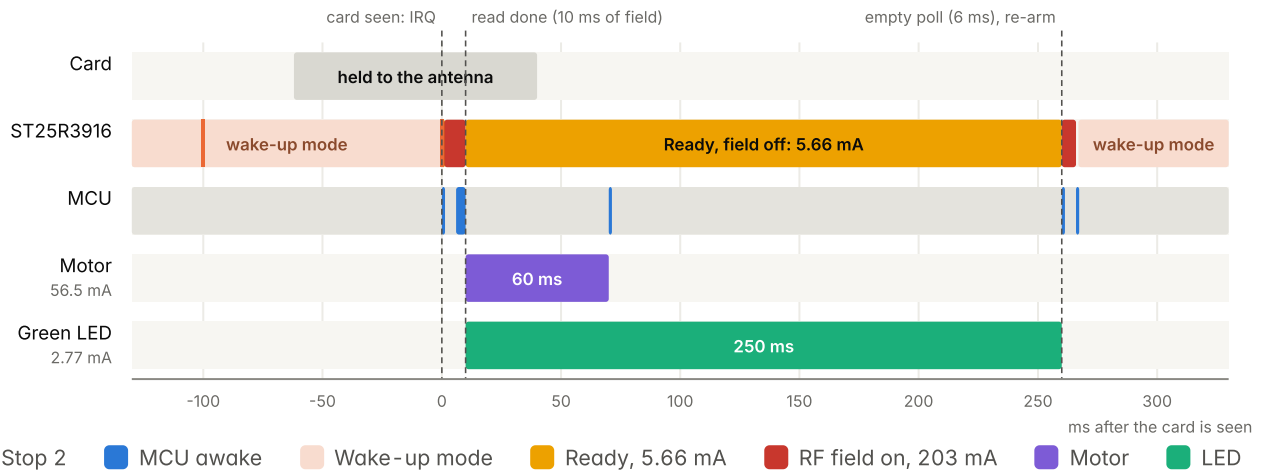
The MCU wakes twice every 5 s: when the heartbeat LED goes on and 10 ms later when it goes off. The battery is sampled on the first of those wakes every 10 s, so it adds no wake of its own. Meanwhile the reader measures the antenna 10 times a second, without the MCU. Nothing else is due, so the loop sleeps the full 5 s (APP_SLEEP_MAX_MS).

The reader's wake-up measurements, every 100 ms



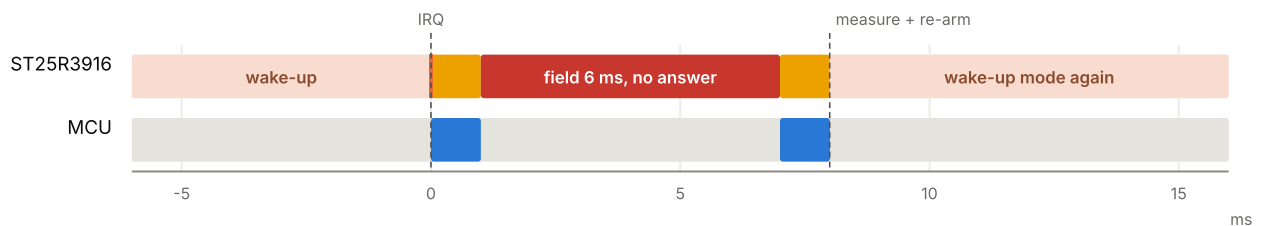
Log scale. Ten times a second the ST25R3916 starts its oscillator, drives the antenna for a moment and compares the amplitude with its reference; the MCU sleeps through all of it. When a card changes the amplitude past the window (the fourth measurement), the reader raises its IRQ and the MCU wakes to read the card. The charge per measurement, 18.5 μ C, follows from the measured 185 μ A average; how it divides between oscillator start-up and the field burst is an illustration (assumed 1 ms start-up) until the pulse is captured with a scope across a shunt resistor.

One accepted card tap



The card is noticed at the reader's next measurement, at most 100 ms after it arrives. The IRQ wakes the MCU, which turns the field on and sleeps through the 6 ms guard in Stop 2, then runs REQA, anticollision and SELECT (~4 ms), logs the card and starts the pattern. The reader stays in Ready mode while the pattern plays, so the motor never runs with the field on. At the end of the pattern one empty poll (the card has gone) re-arms wake-up mode. Five seconds after the last tap the MCU wakes once more to write the RAM buffer to flash.

A false wake-up



A measurement strays past the window with no card there (noise, a hand, drift). One empty poll, about 6 ms of field, and the reader is armed again with a fresh reference: 1.3 mC each, against 5.5 mC with the three empty polls used before.

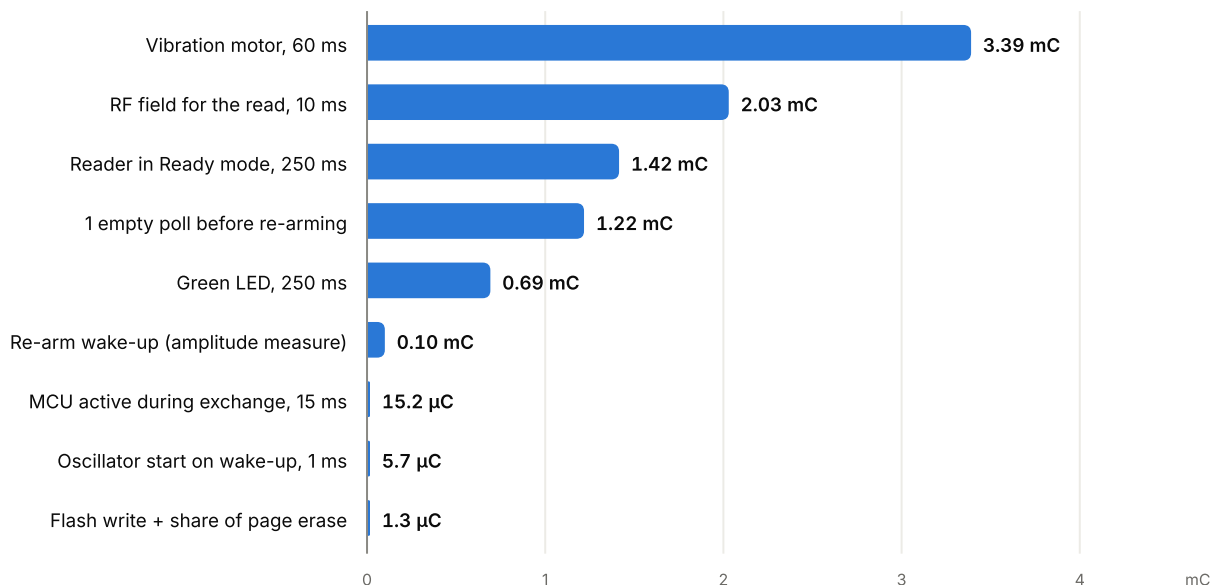
6 Charge per event

Charge = current × time for each load, summed over the event. 1 μ Ah = 3.6 mC.

Event	What draws current	Charge (typ.)	μ Ah (typ.)	μ Ah (pess.)
Accepted card tap	Field 10 ms, motor 60 ms, LED 250 ms, Ready 250 ms, 1 empty poll	8.87 mC	2.46	2.80
Duplicate tap	Same read, 2 × 60 ms motor + LED	11.67 mC	3.24	3.58
False wake-up	Oscillator start + 1 empty poll (6 ms field), no feedback	1.33 mC	0.37	0.48
Power-on (button)	Boot (20 ms at 80 MHz), reader start, 120 ms motor, 300 ms LED	7.97 mC	2.21	2.21
Lecture start (2-5 s hold)	3 s hold, 60 ms + 3 × 80 ms motor, 880 ms LED, marker write	20.63 mC	5.73	5.73
Auto power-off (3 min idle)	250 ms motor, 700 ms red LED, flush, reader power-down	16.23 mC	4.51	4.51
Power-off (5 s hold)	As above + 5 s hold + 60 ms hold buzz	21.67 mC	6.02	6.02

Event	What draws current	Charge (typ.)	μAh (typ.)	μAh (pess.)
Short press (battery check)	0.2 s press, two 150 ms green blinks	0.91 mC	0.25	0.25
Flash page erase	22 ms at ~7 mA, once per 254 records	0.15 mC	0.04	0.04
Battery sample	1.5 ms MCU + ADC, every 10 s, on a heartbeat wake	1.8 μC	0.0005	0.0005

One accepted card tap, 8.9 mC (2.46 μAh)



The field (read plus one empty poll) and the motor are about 75 % of a tap. Keeping the reader in Ready mode during the pattern is most of the rest.

Motor	= 56.5 mA × 60 ms	= 3.39 mC
Ready mode	= 5.66 mA × 250 ms (pattern)	= 1.42 mC
Empty poll	= 1 × 203 mA × 6 ms	= 1.22 mC
Read	= 203 mA × 10 ms (guard, REQA, anticollision, SELECT)	= 2.03 mC
LED	= 2.77 mA × 250 ms	= 0.69 mC
Total	= incl. re-arm, MCU, oscillator, flash	= 8.87 mC = 2.46 μAh

7 A lecture, and battery life

The model lecture: the lecturer powers the unit on and starts a lecture with a button hold; 100 students tap over 15 minutes, 5 of them twice; 3 minutes after the last tap the unit switches itself off. It is on for 18 minutes.

Part	μAh	Share
100 accepted taps	246.3	72 %
5 duplicate taps	16.2	5 %
False wake-ups (1/min)	6.6	2 %
Idle scanning, 18 min	58.2	17 %
Lecture start (button hold)	5.7	2 %
Auto power-off pattern	4.5	1 %

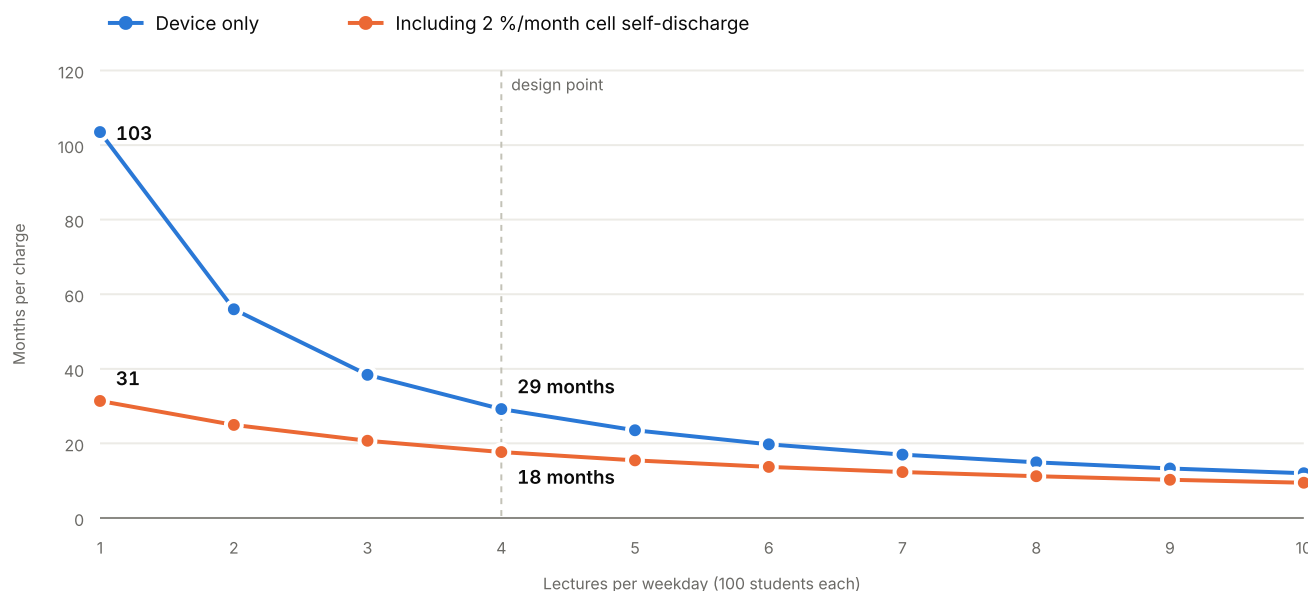
Per month at 4 lectures/weekday:

lectures	= 4 × 21.7	= 86.9
device	= 86.9 × 0.340 mAh	
	+ 1.80 μA × 730 h	= 30.8 mAh
self-dis.	= 2 % × 1000 mAh	= 20.0 mAh
total		= 50.8 mAh

Life = 900 mAh / 50.8 mAh = 17.7 months
(device only: 29.2 months)

Part	μAh	Share
Power-on	2.2	1 %
Total per lecture	340	100 %

Months per charge against lectures per day



Light use is limited by the cell's self-discharge, not the device. At 4 lectures a day a charge lasts about 18 months, with idle scanning and false wake-ups a large share of each lecture.

Scenario (weekdays)	μAh per lecture	Months, device only	Months, with self-discharge	Months, pessimistic
Light: 1/day × 50 students	207	155	35	23
Typical: 4/day × 100 students	340	29	18	13
Busy: 6/day × 150 students	473	14	11	8
Heavy: 8/day × 200 students	602	8	7	6

Shelf time when never switched on: 900 mAh / 1.8 μA = 57 years from the device's draw alone, but self-discharge empties the cell in roughly 3.8 years. Store the device charged and recharge it each term. All values are at room temperature; Li-ion capacity falls 10-20 % near 0 °C and with ageing, partly covered by the usable-capacity factor.

8 Card reads per charge

A tap alone costs 2.46 μAh, so a charge could fund **365,374** taps if nothing else drew current. With each lecture's idle time, power-on and power-off, duplicates and false wake-ups shared out among its students, the effective cost per student is higher, especially in small classes:

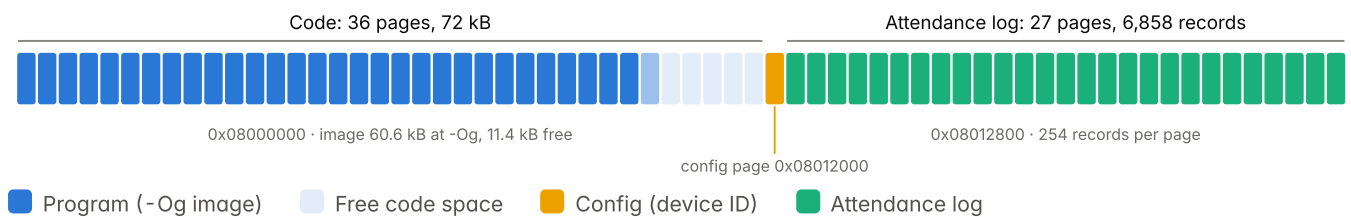
Usage pattern	µAh per student (all overheads)	Taps per charge (900 mAh)	Flash fills per charge
30 students per lecture	5.25	171,282	25x
60 students per lecture	3.91	230,009	34x
100 students per lecture	3.40	264,873	39x
200 students per lecture	3.01	298,846	44x

Even small classes get over 171 000 reads per charge (self-discharge aside), which is 25 to 44 times the 6,858 records the flash can hold. Before the battery is a limit, the log has to be read and cleared by the PC app many times over (by #CLEARLOG when it starts a lecture).

9 Flash: program and attendance log

The fitted MCU is the **STM32L432KB**: its flash - size word at 0x1FFF75E0 reads 128 (measured on the board). Its 128 kB is 64 pages of 2 kB, erased a page at a time, and the program and the log must share them. Earlier firmware assumed the 256 kB STM32L432KC and kept its data at 0x08020000, which the KB does not specify; the layout below fits the KB.

Flash map, one block per 2 kB page

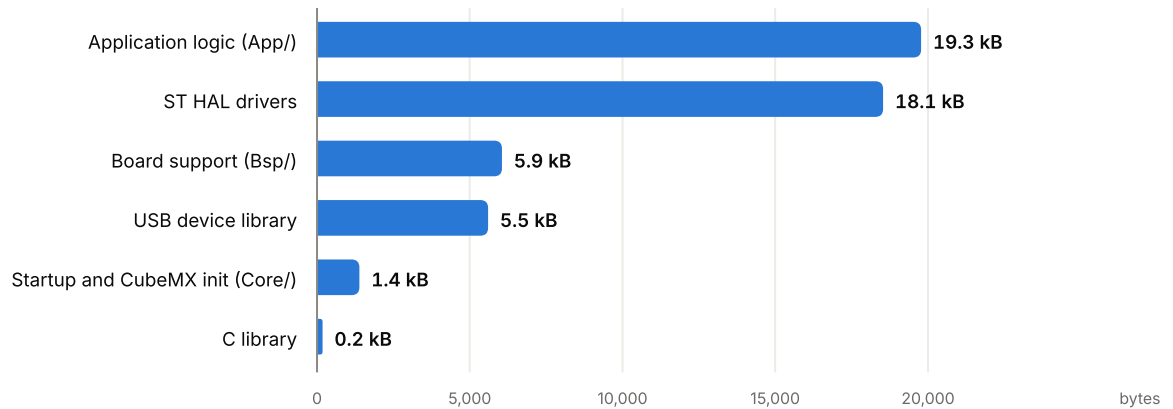


Program size

Measured with the project's own toolchain (arm-none-eabi-gcc 13.3, nano libc, unused sections removed):

Build	Image (bytes)	In 72 kB	Fits	Note
CubeIDE Debug before this change (-OO)	99,244	96.9 kB	No	Larger than any split that leaves a useful log
make and CubeIDE Debug (-Og)	62,052	11.4 kB free	Yes	84 % of the code region; full debugging
CubeIDE Release (-Os)	51,496	21.7 kB free	Yes	Shipping build
-Os with link-time optimisation	46,220	26.9 kB free	Yes	Not used: hardest to debug

Where the 50.3 kB -Os image goes



Application code and ST's HAL are about three quarters of the image. The HAL is used through its generic drivers; replacing it with direct register code could save several kB more, at the cost of portability.

Choosing the split

Every page given to code is 254 records fewer in the log. The CubeIDE Debug build used to compile at -O0 (97 kB), which fits no useful split, so it now builds at -Og like the Makefile. With -Og as the largest build that must fit:

Code / data	Log pages	Records	Lectures of 100	Headroom at -Og	Headroom at -Os
64 kB / 64 kB	31	7,874	74	3.4 kB	13.7 kB
72 kB / 56 kB (chosen)	27	6,858	64	11.4 kB	21.7 kB
80 kB / 48 kB	23	5,842	55	19.4 kB	29.7 kB

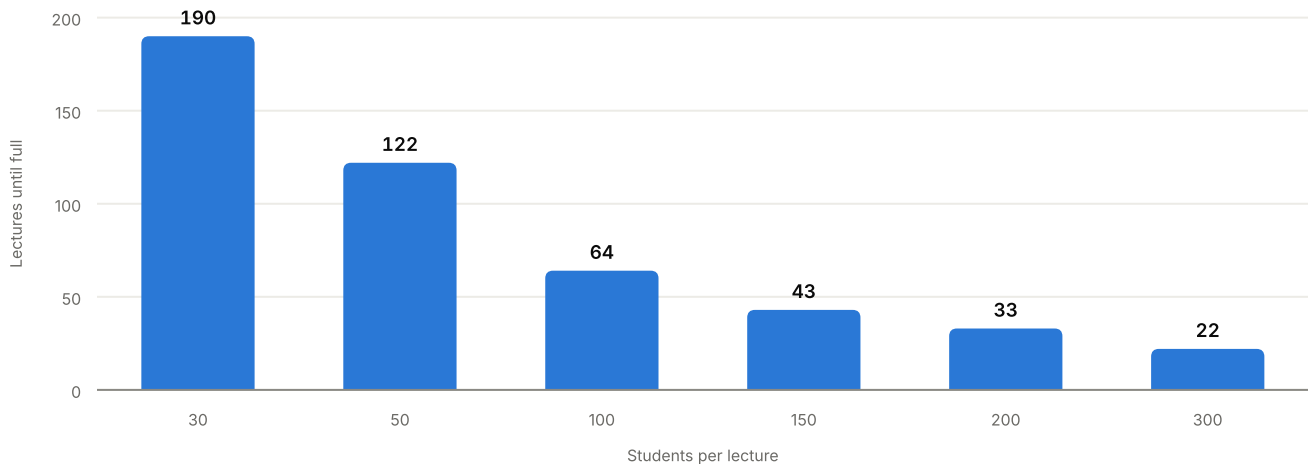
72 kB of code leaves 11.4 kB (16 %) for future features at -Og and still gives the log 6,858 records. A 64 kB code region would add 1,016 records (+15 %) but leave only 3.4 kB at -Og, forcing -Os for debugging. If records ever matter more than debugging, -Os with link-time optimisation (45.1 kB) would make the 64 kB split comfortable.

The linker script (STM32L432KBUX_FLASH.ld) gives the program 72 kB, so an image that outgrows it fails to link instead of overwriting the log. Link-time and compile-time checks keep the linker script, the board header and the log layout in step.

Log capacity

```
Page      = 2048 bytes = 256 double-words
Records   = 256 - 1 header - 1 footer           = 254 per page
Log        = 27 pages × 254                     = 6,858 records (8 bytes each: card ID + time)
Lecture    = 1 header + ceil((module + lecture + 2 NULs) / 4) text records
            = 1 + ceil((6 + 10 + 2) / 4) = 6 for "EE4020" / "Lecture 12" (range 4-16)
Lectures   = floor(6,858 / (N students + 6))
```

Lectures the log can hold before a PC sync



Duplicates are not stored, so N counts distinct students in each lecture.

Students / lecture	Records / lecture	Lectures until full	Student taps stored	At 4 lectures per weekday
30	36	190	5,700	48 days (9.5 weeks)
50	56	122	6,100	30 days (6.1 weeks)
100	106	64	6,400	16 days (3.2 weeks)
150	156	43	6,450	11 days (2.1 weeks)
200	206	33	6,600	8 days (1.6 weeks)
300	306	22	6,600	6 days (1.1 weeks)

- **Students per lecture:** up to **6,852** in a single lecture after a clear (log size minus one marker).
- **Registered students:** not limited by the device. It keeps no card list; any 32-bit card ID below 0xFFFFFFFF00 is stored, and the PC app decides who each card belongs to.
- **When the log is full** new taps are refused with the red error pattern, because APP_LOG_WRAP_WHEN_FULL is 0. Up to 128 records wait in RAM (flushed at 80 % or 5 s after the last tap) and are written before power-off.
- **USB view:** a full log is 6,858 rows × 32 bytes = 219 kB of ATTEND.CSV (429 clusters), well within FAT12's 4084-cluster limit.
- **Flash wear:** each log page is erased once per fill-and-clear cycle. At the STM32L4's 10,000-cycle endurance, that is 10,000 full logs (about 69 million records): not a limit in practice.
- **Moving a unit to this layout:** the new firmware does not read the old data at 0x08020000. Import the unit's log with the PC app before reflashing, and set its device ID again afterwards.

10 Where to save more power

Ranked by effect. The reader's wake-up mode dominates idle current, so its period and the antenna tuning matter most; the rest are small.

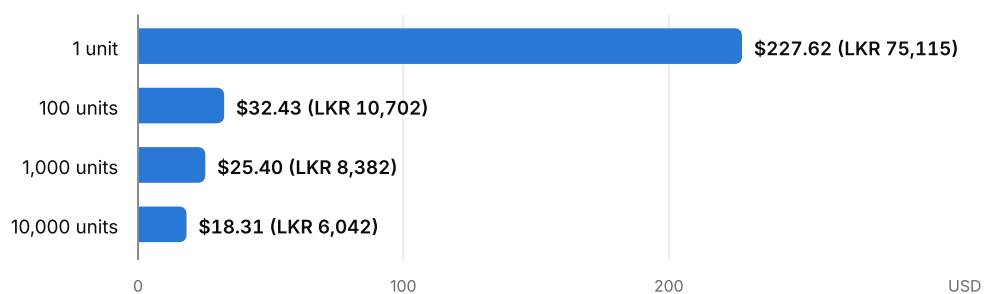
Change	Saving	Effect
Wake-up period 100 ms to 200 ms	- 92 μ A idle	48 % of idle; a card is noticed up to 100 ms later
Wake-up period 100 ms to 400 ms	- 139 μ A idle	71 % of idle; up to 400 ms to notice a card
Tune the antenna: field 203 mA to ~100 mA	- 1.6 mC per tap	19 % of each tap; probably cheaper wake-up measurements too
SPI prescaler /8 to /2	- 0.28 mC per read	3 % of each tap
Motor pulse 60 ms to 40 ms	- 1.1 mC per tap	13 % of each tap
Heartbeat off	- 5.5 μ A idle	3 % of idle; no sign the unit is on

11 Manufacturing cost

What a finished unit costs to make and land in Sri Lanka: the assembled board, the 1000 mAh cell, the motor, the case and box, freight from China, import charges, and final assembly and test in Sri Lanka. Prices are in US dollars with rupees at 1 USD = 330 LKR. The model is `Firmware/docs/cost_model.py`; it reads the LCSC BOM workbook in `PCB/bom`.

USD unless stated	1 unit	100 units	1 000 units	10 000 units
Electronic parts (LCSC)	53.97	1,283	11,197	98,380
PCB fabrication	2.00	30.00	240.00	1,800
PCB assembly (JLCPCB)	76.13	146.61	716.37	5,228
Battery, motor, case, box	16.00	895.00	6,700	34,500
Freight to Colombo	30.00	60.00	350.00	1,500
Import duty, VAT, levies (25 %)	44.52	603.69	4,801	35,352
Final assembly and test (Sri Lanka)	5.00	62.50	633.33	2,683
Yield loss	-	162.16	761.97	3,662
Order total, USD	227.62	3,243	25,399	183,106
Order total, LKR	75,115	1,070,226	8,381,667	60,424,952
Cost per unit, USD	227.62	32.43	25.40	18.31
Cost per unit, LKR	75,115	10,702	8,382	6,042
Price per unit at 40 % margin, USD	379.37	54.05	42.33	30.52
Price per unit at 40 % margin, LKR	125,192	17,837	13,969	10,071

Cost per unit against order size



A single unit carries a whole minimum order: 5 PCBs, 2 assembled boards, JLCPCB's setup and 20 Extended-part fees (\$61), and a courier. From 100 units the parts dominate, led by the STM32 and the ST25R3916.

Where the money goes

- **1 unit:** assembly fees (\$76 for the order) and parts in minimum lots (\$27 a board). Hand-soldering the prototype saves the assembly fees.
- **100 units:** parts are \$12.83 a board; the two ICs are over half.
- **1 000 units:** the case tooling adds \$3.00 a unit; parts fall to \$11.20.
- **10 000 units:** parts \$9.84; import charges are the second largest line.
- **Selling price** = cost / (1 - margin); the table shows a 40 % gross margin as an example.

Key part prices

Part (USD each)	1	100	1 000	10 000
STM32L432KBU6 (U1)	5.075	3.530	3.247	2.950*
ST25R3916-AQWT (U5)	5.440	3.753	3.400*	3.000*
MCP73833 charger (U2)	1.902	1.240	1.118	1.000*
TPS7A0233 LDO (U3)	0.557	0.360	0.335	0.300*
USB4110 USB-C (J1)	1.396	0.824	0.720*	0.620*
PTS636 switch, each (SW1-3)	0.356	0.257	0.187	0.177
NTC 10k (TH1)	0.626	0.385	0.311	0.280*
All electronic parts, per board	26.98	12.83	11.20	9.84

* extrapolated (LCSC lists no break at that quantity).

Assumptions

Item	1 / 100 / 1 000 / 10 000	Basis
Exchange rate	1 USD = 330 LKR	Rate on 3 October 2026; it moved 328-332 over the month before
Electronic parts	LCSC, 8 October 2026	Price breaks above; other parts at the BOM's minimum-lot price × 1, 0.85, 0.65, 0.5
PCB	\$2 for 5; then \$0.30 / \$0.24 / \$0.18	2 layers, 60 × 89 mm, per board (estimate)
Assembly	JLCPCB Economic to 100, Standard above	265 SMT + 18 hand-soldered joints, 20 Extended part types at \$3.07 each per order
Battery	\$6.00 / \$3.00 / \$2.20 / \$1.80	1000 mAh LiPo with protection board and JST XA lead (estimate)
Motor	\$1.00 / \$0.45 / \$0.30 / \$0.25	10 mm coin ERM with leads (estimate)
Case	\$8.00 / \$5.00 / \$3.90 / \$1.20	3D printed to 100; aluminium tool (\$3 000) at 1 000; steel tool (\$6 000) at 10 000
Freight	\$30 / \$60 / \$350 / \$1,500	Whole order, China to Colombo; lithium cells ship as dangerous goods
Import charges	25 % of CIF	Duty (HS code dependent) + 18 % VAT + 2.5 % SSCL + levies; VAT is reclaimable when registered
Labour	\$2.50/h; 120 min / 15 min / 8 min / 5 min	Fit battery, motor and case, flash, test; plus a \$300-600 test jig from 1 000

Item	1 / 100 / 1 000 / 10 000	Basis
Yield	100 % / 95 % / 97 % / 98 %	Units that pass test

Risks and exclusions. LCSC held only 27 ST25R3916 on 8 October 2026 and Digi-Key none, so 100 units or more need a quote from ST or a distributor. The duty rate depends on the HS code the goods are cleared under. Not included: the optional PN532 module, a USB cable, radio type approval (TRCSL, for a 13.56 MHz transmitter), firmware and engineering time, warranty and distribution. R6 is still 1 k Ω (1 A charge current); the BOM's 2 k Ω alternate (500 mA) suits a USB-C sink and costs the same.

Sources: lcsc.com product pages C2928224, C2908147, C627668, C2887324, C5143397, C2689642, C141588 (8 October 2026); jlcpcb.com/help/article/pcb-assembly-price; jlcpcb.com/news/discounts-for-large-pcb-orders; exchange-rates.org (USD/LKR); customs.gov.lk "Computation of import levies" (2026); KPMG, Sri Lanka budget 2026 tax proposals; teamrapidtooling.com (mould costs); besomi.com (603450 LiPo retail price).

12 Verification plan

- Fit and tune the antenna (C16-C19, R15), then repeat modes 2 and 4 of the power-test build: the field current and the wake-up mode average should both fall.
- Read dbg_nfc_false_wakes after an hour of idle with no card near (plug in USB to keep the MCU awake for the probe) and replace the assumed 1 per minute.
- Integrate one tap and one idle minute with a current-integrating meter (Nordic PPK2 or similar), with no debugger attached, and compare with 8.9 mC and 194 μ A.